

OMID ZANGANEH

Senior GIS Developer and Analyst

MS Geography — Geographic Information Science & Technology · Workflow Automation · AI/ML Integration
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PROFESSIONAL SUMMARY

GIS Developer and Analyst who builds production automation for telecom and fiber engineering. At Olsson, ships Python and C# apps, ArcGIS Pro add-ins, and geoprocessing toolboxes that cut multi-day GIS work to minutes — including network design, expansion planning, and multi-source data acquisition. Also builds AI tools (Azure OpenAI, Gemini, YOLO) that reduce contract sourcing from months to hours. 2025 Edison Award Nominee and 2026 Edison Award Winner at Olsson. MS Geography (GIS&T), 4.0 GPA.

WORK EXPERIENCE

Olsson ♦ 2025 Nominee & 2026 Edison Award Winner

March 2025 – Present

GIS Developer and Analyst
Lincoln, Nebraska

- Built production Python and C# GIS apps, including bore profile automation that cut processing from days to minutes.
- Created an ArcGIS Automation Suite of Python toolboxes that reduced manual GIS steps by ~90% and sped up fiber network design.
- Shipped ArcGIS Pro add-ins (.NET/C#/WPF): GIS Data Downloader, FTTH Network Designer, RF Analysis panel, and Street View map tool.
- Built AI tools with Azure OpenAI and Google AI Studio — including RFP Radar — cutting sourcing timelines from months to hours; plus YOLO apps for utility infrastructure inventory.

University of Nebraska at Omaha

January 2024 – August 2025

Graduate Teaching Assistant — Instructor of Record
Omaha, Nebraska

- Taught Human-Environment Geography labs as sole instructor of record to 150+ students across three semesters.

University of Nebraska at Omaha

June 2024 – August 2025

GIS Technician — Omaha Spatial Justice Project
Omaha, Nebraska

- Digitized historical parcels and legal records into a geodatabase of racially restrictive covenants in Douglas County.

SELECTED PROJECTS

RFP Radar — AI-Powered RFP Sourcing

Azure OpenAI · Google Gemini · Playwright · Python/C#

Desktop app for web-grounded RFP search, classification, and CSV export. Months of sourcing → hours.

Bore Profile Automation — Directional Drilling Profile Generator

Python · C# · ArcGIS Pro · SQL Server · Matplotlib

Reads spatial waypoints and elevation models; generates 2D/3D bore profiles. Days → minutes.

Fiber Automatic Expansion — Automated Fiber Build-Area Planning

C# · .NET · ArcGIS Pro SDK · WPF · Spatial Analysis

Grids the study area for density analysis, flood-fills viable build zones by PPM thresholds, and generates economic scorecards with live refinement. Days/weeks → minutes.

FTTH Network Designer — Automated Fiber Optic Network Planning

C# · .NET 8 · ArcGIS Pro SDK · WPF · Kruskal MST

Places shafts, connects homes, and builds an optimal Main Trunk / Terminal Branch backbone. Days → minutes.

ArcGIS Data Downloader Add-in — Multi-Source GIS Data Acquisition

C# · .NET 8 · ArcGIS Pro SDK · WPF · REST APIs

Downloads OSM, USGS, FEMA, Census/TIGER, Wikipedia, and BSL layers into ArcGIS Pro.

RF Analysis Tool — 8-Tool Wireless Planning Panel

C# · ArcGIS Pro SDK · Python

Coverage prediction, PCI/RSI planner, tilt/azimuth optimizers, interference analysis, and tower placement — all as map layers.

Aerial & Streetview AI Detection — YOLO Utility Infrastructure Detection

Python · YOLO · OpenCV · Aerial/Street Imagery APIs

Detects and classifies utility assets from aerial and street imagery; exports georeferenced results to ArcGIS.

GeoPipe — Enterprise GIS ETL to SQL Server

Python · CustomTkinter · pyodbc · SQL Server · PyInstaller

GUI ETL for large spatial/tabular imports with schema auto-detection, GEOMETRY/GEOGRAPHY support, and connection-loss auto-resume.

omidzanganeh.com — Full-Stack Portfolio & Browser GIS Tools

Next.js 16 · React · TypeScript · Leaflet · Vercel

Personal site with 10+ browser GIS tools (geocoder, isochrone, elevation, census, coordinate converter), Gym Flow, Stock Screener, and an AI news aggregator.

EDUCATION

Master of Science, Geography — Geographic Information Science & Technology August 2025

University of Nebraska at Omaha, Nebraska **GPA: 4.00** · **GRACA Project Award**

Thesis: Spatiotemporal Analysis of NOx Emissions from U.S. Cement Plants Using TROPOMI Data — remote sensing, hotspot analysis, and environmental justice.

Bachelor of Science, Geomatics (Surveying) Engineering

August 2016

Geomatics College of National Cartographic Center (GCNCC), Tehran

TECHNICAL SKILLS

Languages & Frameworks Python · C# / .NET · SQL · TypeScript · JavaScript · HTML/CSS · React · Next.js · WPF

GIS & Spatial ArcGIS Pro · ArcGIS Pro SDK · ArcPy · Python Toolboxes (.pyt) · ArcGIS Online · ArcGIS Enterprise · QGIS · Geoprocessing · Network Analysis · Remote Sensing · Google Earth Engine

AI / ML Azure OpenAI · Azure AI Foundry · Google AI Studio / Gemini · YOLO · OpenCV · Prompt Engineering · Web Grounding · Batch Classification Pipelines

Data & Backend SQL Server · Supabase · PostgreSQL · pyodbc · ETL Pipelines · Spatial Types (GEOMETRY/GEOGRAPHY) · Smartsheet API

Web & Cloud Next.js App Router · Vite · Leaflet · Vercel · REST APIs · PWA · Microsoft Azure · Google Cloud

Desktop & Tooling CustomTkinter · PyInstaller · Playwright · Matplotlib · AutoCAD · ENVI · SNAP · Photomod · Tableau · Adobe Photoshop · Adobe Illustrator

Languages English (Fluent) · Persian (Native)

For project write-ups, live GIS tools, and demos: omidzanganeh.com
